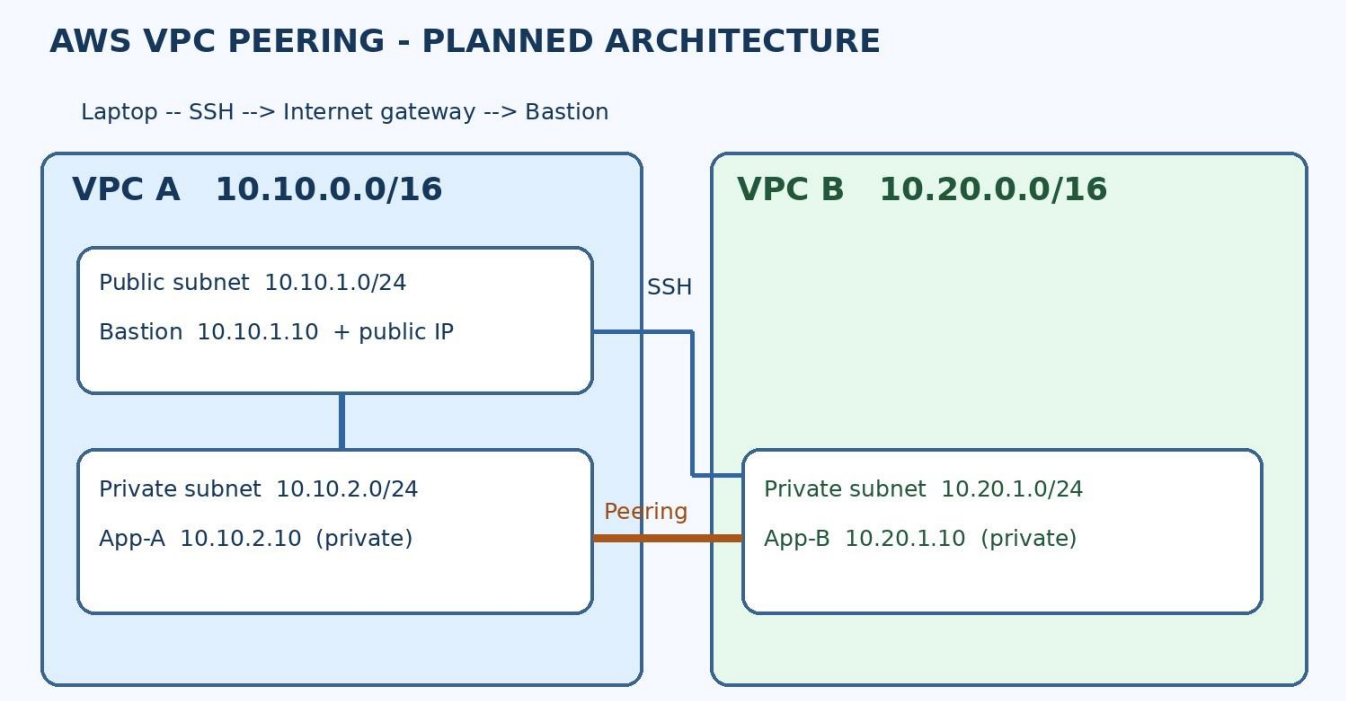
AWS VPC Peering

Objective

Establish a private peering connection between two non-overlapping VPCs, access private hosts through a bastion, verify two-way traffic, and demonstrate a controlled ICMP failure and recovery.

Planned architecture



Resource inventory

Resource	Planned address	Actual AWS ID
VPC A	10.10.0.0/16	[vpc-...]
VPC B	10.20.0.0/16	[vpc-...]
Public A subnet	10.10.1.0/24	[subnet-...]

Resource	Planned address	Actual AWS ID
Private A subnet	10.10.2.0/24	[subnet-...]
Private B subnet	10.20.1.0/24	[subnet-...]
Bastion	10.10.1.10	[i-...]
App-A	10.10.2.10	[i-...]
App-B	10.20.1.10	[i-...]
Peering	A <-> B	[pcx-...]

Configuration summary

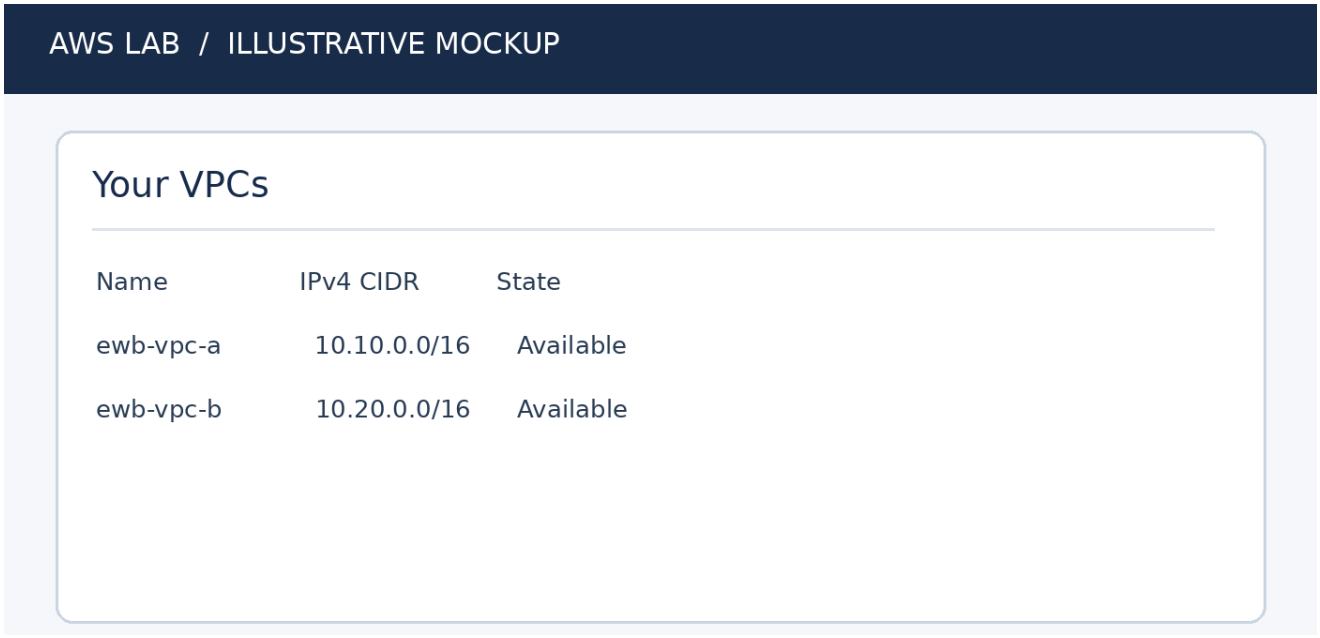
Route tables: VPC A public: local, 0.0.0.0/0 to internet gateway, 10.20.0.0/16 to peering; VPC A private: local and 10.20.0.0/16 to peering; VPC B private: local and 10.10.0.0/16 to peering. Each table must be explicitly associated with its subnet.

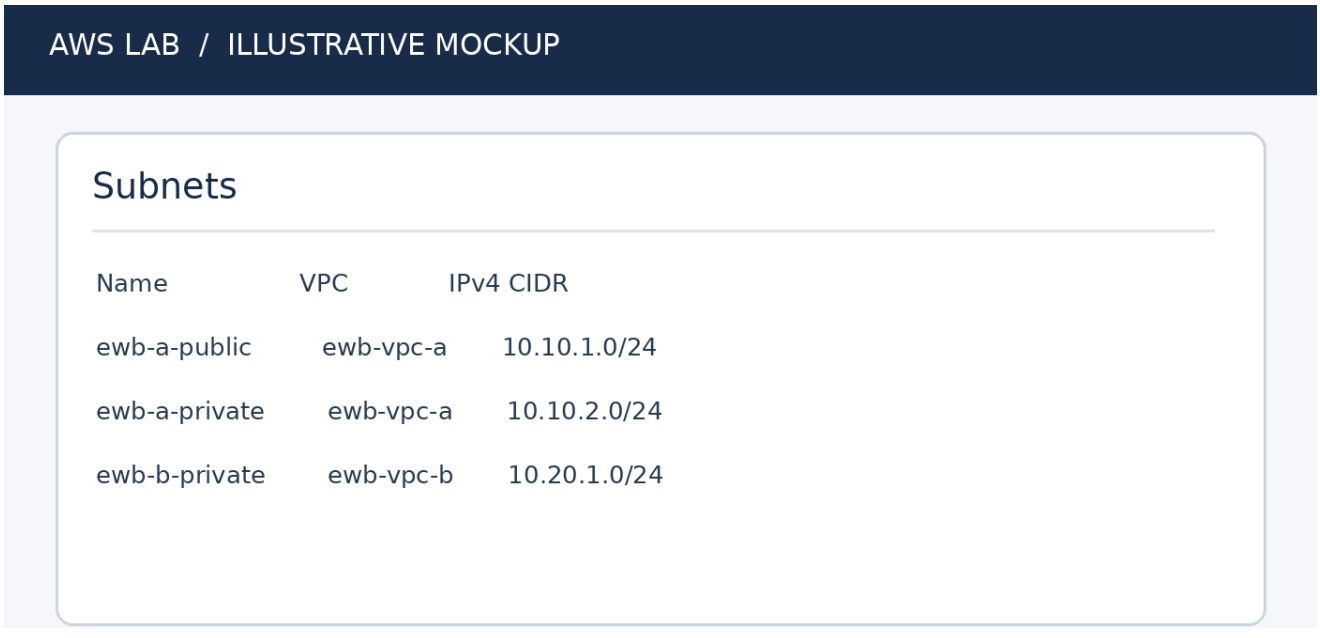
Security: Bastion TCP 22 only from student public /32. App-A TCP 22 from 10.10.1.10/32 and ICMP echo from 10.20.1.10/32. App-B TCP 22 from 10.10.1.10/32 and ICMP echo from 10.10.2.10/32.

SSH: laptop uses ProxyJump via bastion to reach App-A and App-B without placing the private key on the bastion.

Verification evidence - insert real screenshots

01 VPC creation:





AWS LAB / ILLUSTRATIVE MOCKUP

Route tables

ewb-rt-a-public: 10.10.0.0/16 local; 0.0.0.0/0 IGW

10.20.0.0/16 pcx-example

ewb-rt-a-private: 10.10.0.0/16 local; 10.20.0.0/16 pcx-example

ewb-rt-b-private: 10.20.0.0/16 local; 10.10.0.0/16 pcx-example

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Security groups

ewb-sg-bastion	SSH TCP 22 from YOUR_PUBLIC_IP/32
ewb-sg-app-a	SSH TCP 22 from 10.10.1.10/32
ewb-sg-app-a	ICMP Echo Request from 10.20.1.10/32
ewb-sg-app-b	SSH TCP 22 from 10.10.1.10/32
ewb-sg-app-b	ICMP Echo Request from 10.10.2.10/32

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Peering connections

Name: ewb-a-b-peer

Requester: ewb-vpc-a (10.10.0.0/16)

Accepter: ewb-vpc-b (10.20.0.0/16)

Status: Active (illustrative only)

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EC2 instances

ewb-bastion	10.10.1.10	public IPv4: assigned
ewb-app-a	10.10.2.10	public IPv4: none
ewb-app-b	10.20.1.10	public IPv4: none

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Terminal - App-A and App-B

App-A $ ping -c 4 10.20.1.10

4 packets transmitted, 4 received, 0% packet loss

App-B $ ping -c 4 10.10.2.10

4 packets transmitted, 4 received, 0% packet loss

AWS LAB / ILLUSTRATIVE MOCKUP

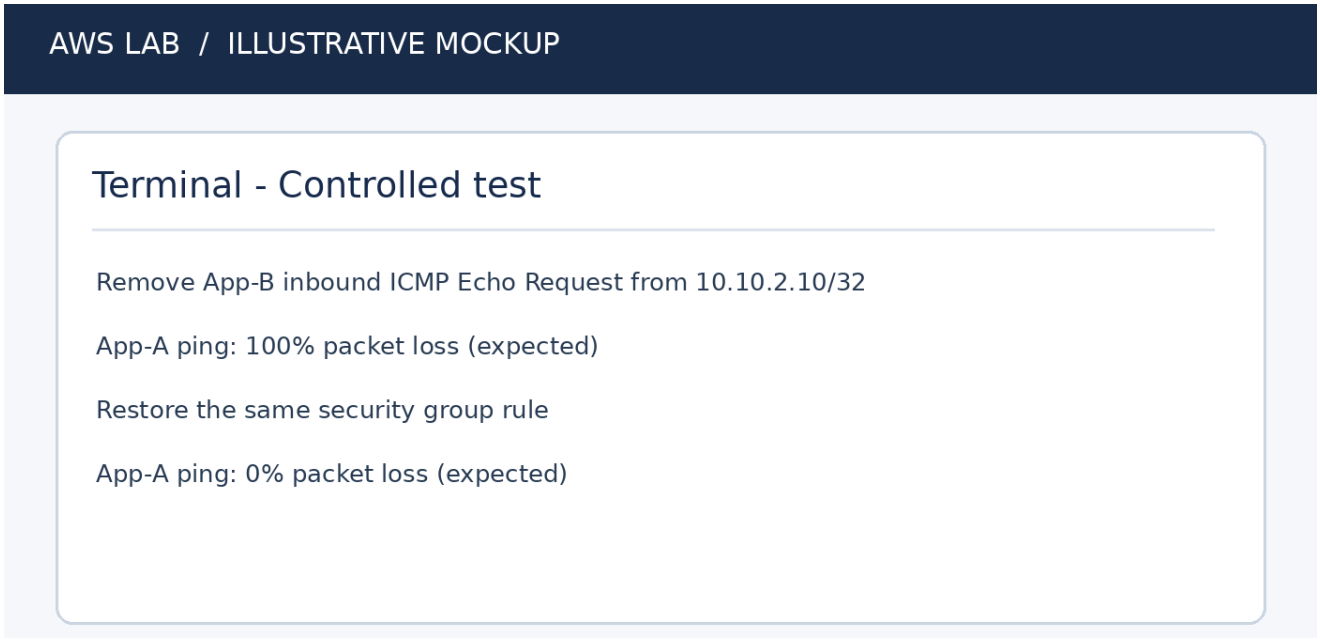
Terminal - App-A - baseline

$ ping -c 4 10.20.1.10

4 packets transmitted, 0 received, 100% packet loss

Expected before peering routes exist





Technical explanation

VPC peering is a private connection between two VPCs with non-overlapping address ranges. An Active peering connection alone is insufficient: route tables must send packets to the peer and the return route must exist. Security groups must allow the intended traffic. VPC peering is non-transitive and does not grant VPC B internet access through VPC A.